

Semiconductor Devices Design and Simulation Portfolio

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COMSOL Simulation Portfolio: https://gla-my.sharepoint.com/:f:/g/personal/3180175o_student_gla_ac_uk/IgBM5yYC4XzAS4hLC3JeDi4DAT7eFPgaLcm0Gdub9iNB4VI?e=echGJh

1 Introduction

Metal-Oxide-Semiconductor Field-Effect Transistors (MOSFETs) are barrier-controlled microelectronic devices; they are important types of transistors that are ubiquitous in integrated circuit technology. They have digital and analogue applications, such as CMOS inverters and common source amplifiers. While there are other effective analogue structures (e.g. bipolar junction transistors), the CMOS remains dominant in digital electronics due to its high input impedance, simple driving, low static power consumption and high device density [?].

To summarise MOSFET operation in one sentence, a gate voltage V_G and drain voltage V_D are applied to control the height and tilt of an energy barrier between the source and drain, controlling the flow of drain current I_D out of the device. The exact physics of the MOSFET is governed by the semiconductor transport equations, and numerical simulations play an important role in the study and design of these devices. COMSOL Multiphysics is an effective software package to this end.

The purpose of device simulation is to predict characteristics and performance, most notably the current-voltage characteristics at the drain contact [2]. This allows manufacturers to prototype devices and understand the effect of design and parameter choices on behaviour building physical devices.

In our work, we use the COMSOL Multiphysics software package to construct a portfolio of silicon-based MOSFETs, simulate their electrostatics, and obtain outputs to measure the device characteristics and performance. We analyse these outputs and compare them to theoretical expectations across parameter choices. We also simulate a p-n junction diode and analyse its characteristics.

2 Simulation Setup

Throughout the COMSOL simulations, we use the drift-diffusion model and Shockley-Read-Hall recombination to compute the electrostatic solutions to the devices. The drift-diffusion model is a model for electron and hole transport within a semiconductor. The drift of carriers is due to the electric field, and the diffusion is due to concentration gradients. Shockley-Read-Hall recombination is a model in which electrons in the conduction band combine with holes in the valence band via intermediate trap states in the bandgap, which are caused by defects and impurities in the semiconductor. For all simulations the semiconductor is silicon and the temperature is 293.15 K.

The doping profile of a silicon-based n-type MOSFET in enhancement mode (E-NMOS) is shown in Figure 2, it consists of a $3 \times 0.7 \times 0.6 \mu\text{m}$ p-type silicon body doped with an acceptor concentration of $p = 10^{17} \text{ cm}^{-3}$, and n-type silicon regions at the source and drain with donor concentration $n = 10^{20} \text{ cm}^{-3}$ and junction depth $0.25 \mu\text{m}$. The gate is of length $L = 1.5 \mu\text{m}$ and width $W = 0.7 \mu\text{m}$, and presides over an oxide layer of thickness $d_{ox} = 30 \text{ nm}$ and relative permittivity $\epsilon_{ox} = 4.5$. There are ohmic contacts at the gate, source, drain and on the underside of the device. This defines boundary conditions that fix the electrostatic potential at the surface to the specified voltage. The work function of the gate electrode is set to 4.1 V. The mesh used in our simulation, shown in Figure 1, has a maximum element size of 69 nm, triangular mesh on the three metal contacts, and a reverse graded mesh of cuboids in the body, with smallest elements near the drain, gate and source.

The n-type MOSFET in depletion mode (D-NMOS) has a very similar structure to the E-NMOS, except that an additional bridge of n-type doping with donor concentration 10^{18} cm^{-1} is introduced across the gate, as shown in the doping profile in Figure 3. The channel has a vertical junction depth of $0.01 \mu\text{m}$ such that a suitable voltage could deplete the channel.

The p-type MOSFET in enhancement mode (E-PMOS) has the same structure as the n-type MOSFET, however the dopings are swapped such that at the drain and source, $p = 10^{20} \text{ cm}^{-3}$, and in the body, $n = 10^{17} \text{ cm}^{-3}$. Figure 4 shows the doping profile.

The p-n junction diode has a different structure. The body is also a $3 \times 0.7 \times 0.6 \mu\text{m}$ silicon block, but the anode and cathode are modelled as ohmic contacts on each end of the block. The body has p and n-type doping of concentration 10^{16} cm^{-3} in each respective half with junction depths $0.2 \mu\text{m}$, as seen in Figure 6. To simulate this, we use a different mesh - this also has a maximum element size of 69 nm, however it was evenly distributed, as in Figure 5. This was sufficient for the behaviour we sought to demonstrate.

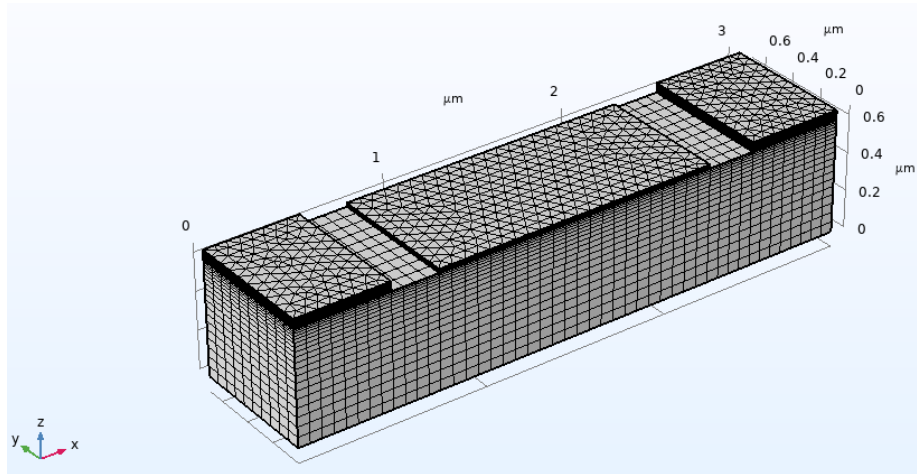


Figure 1: Diagram of mesh used for the numerical simulation of the MOSFETs in COMSOL.

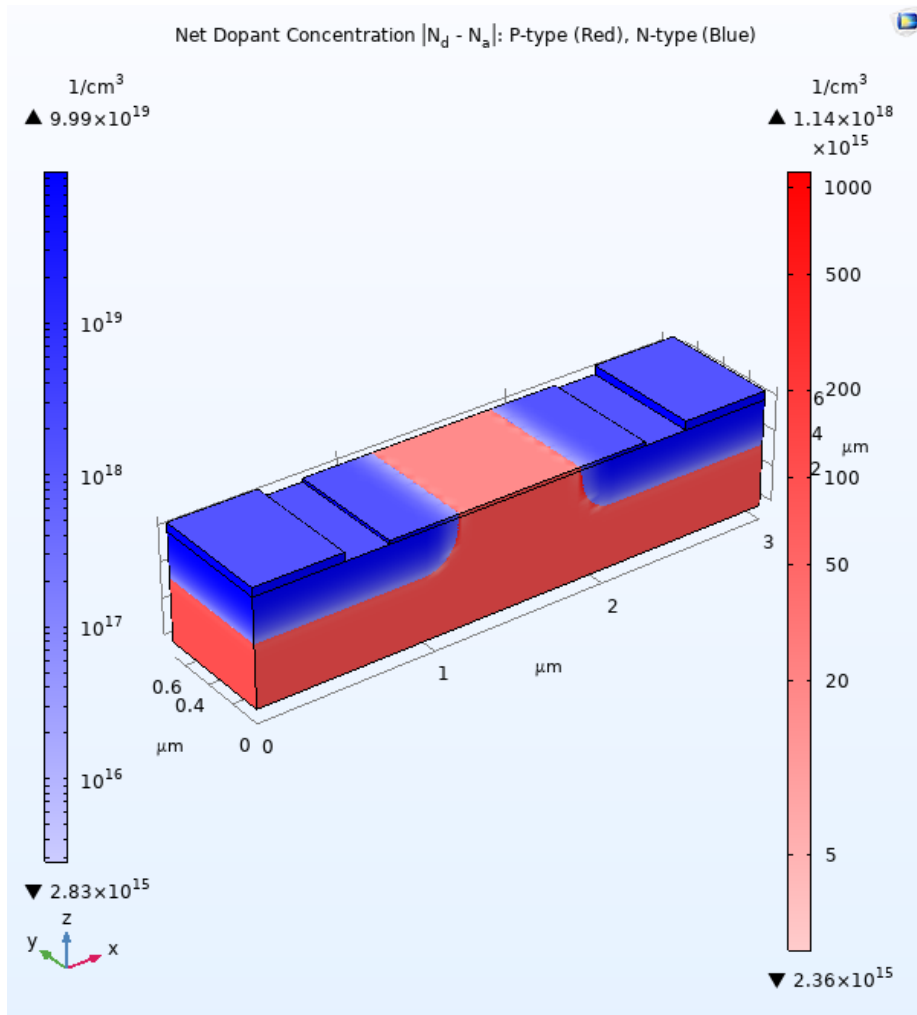


Figure 2: Diagram of the doping profile of the E-NMOS.

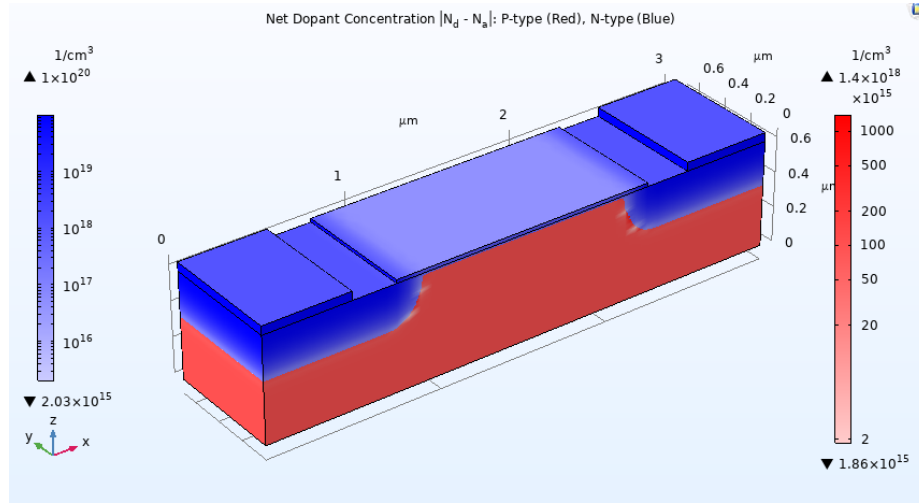


Figure 3: Diagram of the doping profile of the D-NMOS.

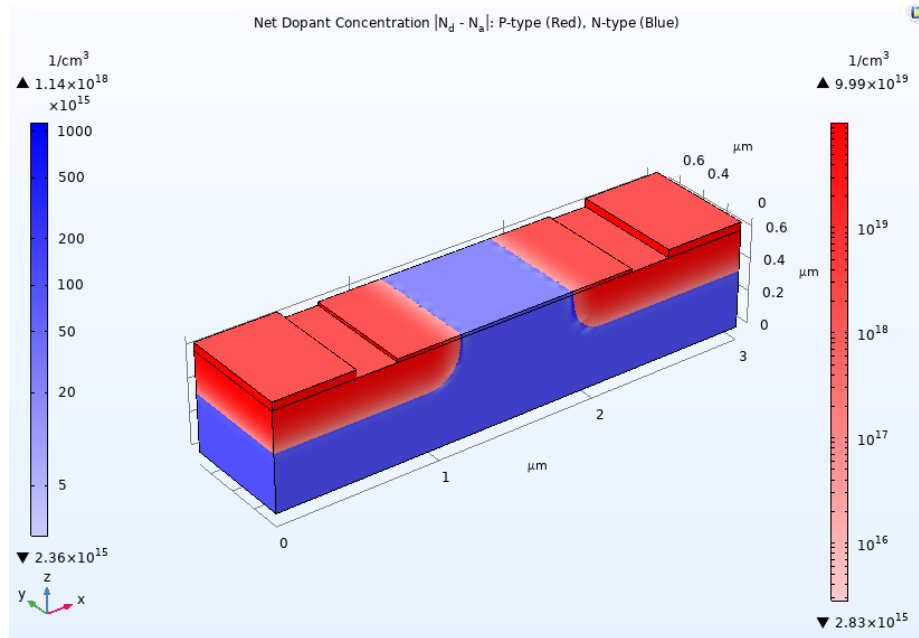


Figure 4: Diagram of the doping profile of the E-PMOS.

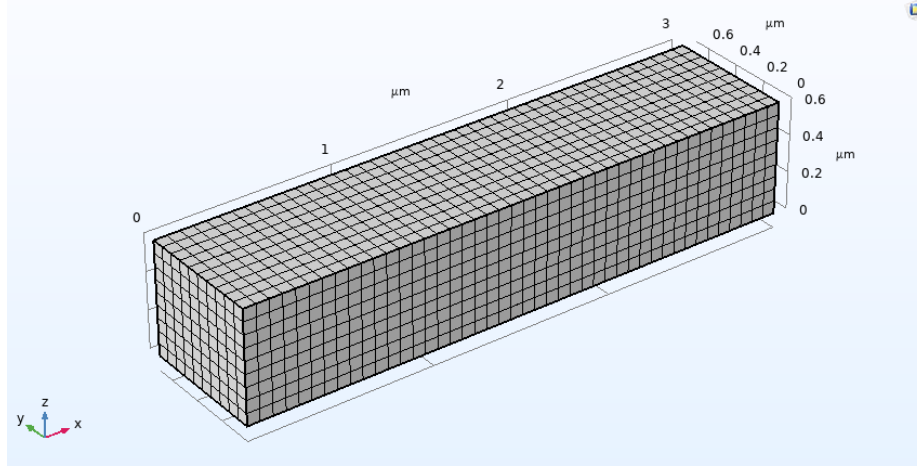


Figure 5: Diagram of mesh used for the numerical simulation of the p-n junction diode in COMSOL.

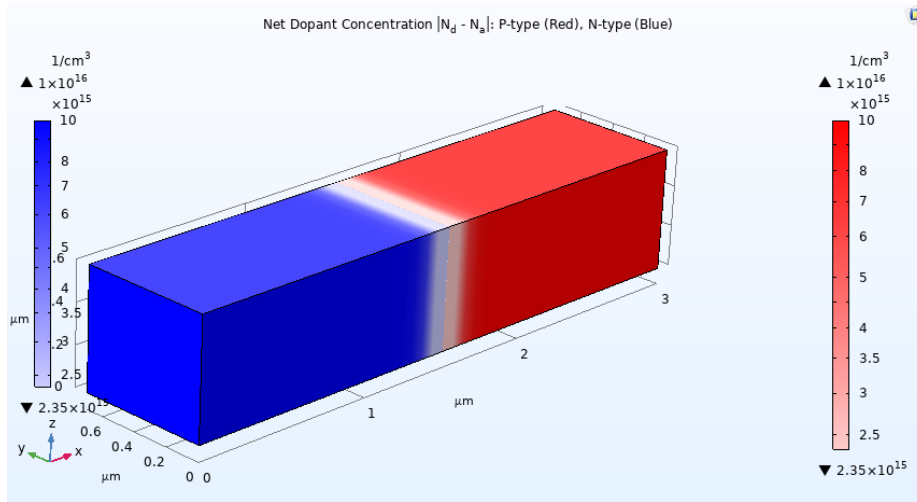


Figure 6: Diagram of the doping profile of the p-n junction.

3 Results and Analysis

3.1 E-NMOS

The behaviour of the E-NMOS is as follows: in the cutoff region ($V_G < V_T$), the channel is depleted; then in the linear region the gate voltage V_G is increased above the threshold voltage V_T and the channel is inverted; lastly in the saturation region, the drain voltage V_D is increased above the overdrive voltage $V_{OV} = V_G - V_T$ and pinchoff occurs in the channel near the drain. These three effects are shown in Figure 7.

These phenomena arise from quantum mechanics, namely the band structure created by the symmetry of the silicon lattice. An effective demonstration of this is the band diagrams vertically through and horizontally along the channel. Firstly, Figure 8 shows the band bending near the gate of the depleted channel ($V_G > V_T = 1.15V$, which we shall see later). Deep in the p-doped bulk, the bands are flat because the electrostatic potential $\phi(x)$ is constant.

$$E_C(\mathbf{x}) = E_C - q\phi(\mathbf{x}) \quad E_V(\mathbf{x}) = E_V - q\phi(\mathbf{x}) \quad (1)$$

The physics underpinning this is that a constant electrostatic potential does not give rise to an electric field $\mathbf{E}$, so electrons and holes in the bulk do not experience a net force $\mathbf{F}$.

$$\mathbf{E} = -\nabla\phi, \quad \mathbf{F} = q\mathbf{E} \quad (2)$$

From this band diagram alone, we can deduce that the bulk is p-doped, this is because the intrinsic energy level E_i is above the Fermi level E_F . This can be seen from the equilibrium equations from statistical physics:

$$p = n_i e^{\frac{E_i - E_F}{k_B T}} > n = n_i e^{\frac{E_F - E_i}{k_B T}} \iff E_i > E_F. \quad (3)$$

Here n_i is the intrinsic carrier concentration, which is equal for holes and electrons.

Near the gate, the positive gate voltage induces an electric field in the semiconductor, and the bands bend downwards as per Equation 1. The quantum mechanics behind this is that the Coulomb force of this field on the electrons lowers the energy of available states. We observe that the conduction band is closer to the electron quasi-Fermi energy level F_n , meaning less energy is required for electrons to excite into the conduction band, and we expect a high population of conduction electrons. This is supported by the Fermi-Dirac distribution for an electron gas [4], which defines the probability that an available state of energy E is occupied by an electron.

$$f(E) = \frac{1}{1 + \exp\left(\frac{E - F_n}{k_B T}\right)} \quad (4)$$

If this electron population exceeds the hole population near the gate, the channel is inverted to n-type semiconductor.

Figure 9 shows the band diagram along the channel under the same voltages as Figure 8. The

flat bands in the drain and source are due to the zero x -component of electric field (note however $\mathbf{E} \circ \mathbf{z} < 0$ in the drain). As per the above, we expect the 5 V drain voltage to lower the energy bands at the drain, and indeed we see a 5 eV = $-qV_D$ drop between the terminals. We note that there is a potential barrier limiting the flow of electrons across the channel (where E_C and F_n split 1.4 μm along the device), and by our previous theory, the height of this barrier is controlled by the gate voltage. Once electrons have overcome this barrier, they experience a Coulomb force down the channel band and into the drain due to the conduction band slope.

$$\begin{aligned} E_C(\mathbf{x}) &= E_C - q\phi(\mathbf{x}), & \mathbf{E} \circ \mathbf{x} &= -\frac{\partial\phi}{\partial x} \\ \implies \frac{\partial E_C(\mathbf{x})}{\partial x} &= q\mathbf{E} \circ \mathbf{x} \implies \mathbf{F} \circ \mathbf{x} &= -\frac{\partial E_C(\mathbf{x})}{\partial x} \end{aligned} \quad (5)$$

The next consideration is the effect of the oxide. This acts as a capacitor and insulator between the gate electrode and semiconductor by creating an electric field and preventing current flow. The gate capacitance C_{ox} is dependent on the gate thickness via the equation

$$C_{ox} = \frac{\epsilon_{ox}}{d_{ox}}. \quad (6)$$

Hence a thinner gate oxide increases the gate capacitance, and in turn lowers the threshold voltage V_T .

$$V_T = V_{FB} + 2\phi_B + \frac{\sqrt{4\epsilon_s qp\phi_B}}{C_{ox}} \quad (7)$$

where V_{FB} is the flatband voltage, ϕ_B is the bulk potential and ϵ_s is the permittivity of silicon [3]. Figure 10 supports this theory, where the threshold voltage (extrapolated root from the linear regime) decreases from 1.15 V to 0.93 V as the oxide thickness d_{ox} decreases from 30 nm to 25 nm.

The saturation region is characterised by a pinching of the inversion channel near the drain when $V_D > V_{OV}$. This is evidenced in the simulation by Figure 11 by the strong electric field for $V_D = 5 \text{ V} > 3 - 1.15 \text{ V}$, as explained by the fluctuation in charge from majority n-doped to p-doped to n-doped across a short distance.

Lastly, we combine our theory into the IV characteristic curve of Figure 12. This shows three plots in inversion with clear linear and saturation regions. In an ideal transistor, the curve in the saturation region would be constant, corresponding to infinite output resistance.

$$r_0 = \left(\frac{\partial I_D}{\partial V_D} \right)^{-1} \Big|_{V_G} \quad (8)$$

In the reality of our device, operating at $V_G = 3 \text{ V}$ we have that $r_0 = 8.1 \times 10^4 \Omega$. Output resistance is an important metric for the performance of a MOSFET as an amplifier, as given by the maximum amplifier gain

$$A_v(max) = -g_m r_0 \quad (9)$$

where g_m is the transconductance.

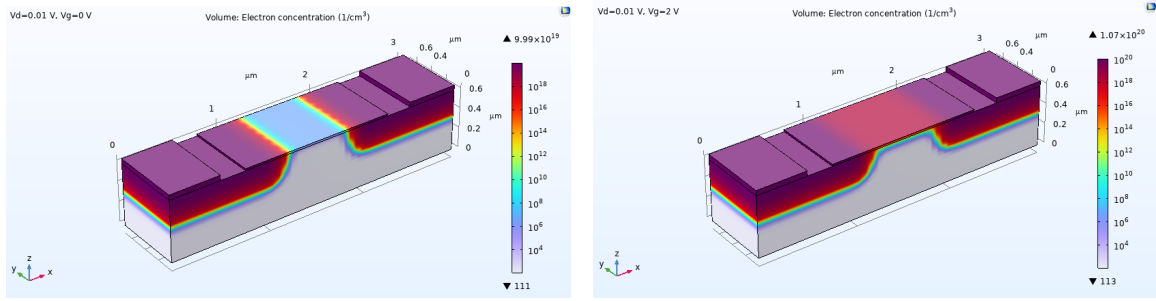
At $V_G = V_D = 3\text{ V}$, the on current is $I_{ON} = 0.24\text{ mA}$. Numerical simulations gave an off current at $V_D = 3\text{ V}$ on the order of 10^{-16} A , which is negligibly small, and I anticipate is unreliable due to the computational limitations of the simulation. These are two significant metrics for digital applications because in a CMOS inverter the switching time is

$$\tau \propto I_{ON}^{-1} \quad (10)$$

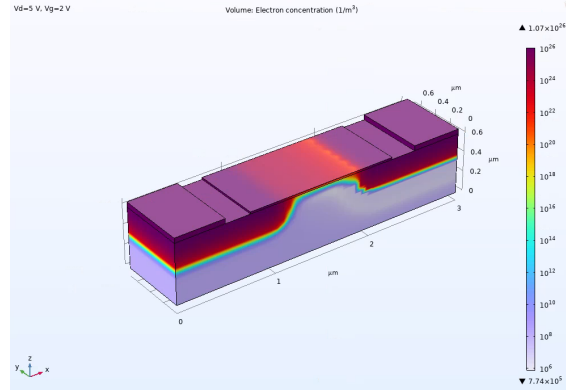
and the static power consumption at operating voltage V is

$$P_{OFF} = I_{OFF}V. \quad (11)$$

Therefore high I_{ON} and low I_{OFF} are desirable characteristics for fast computation and low energy cost.



a) Depletion channel for $V_G = 0\text{ V}$, $V_D = 10\text{ mV}$. b) Inversion channel for $V_G = 2\text{ V}$, $V_D = 10\text{ mV}$.



c) Pinched channel for $V_G = 2\text{ V}$, $V_D = 5\text{ V}$.

Figure 7: 3D plots of the electron concentration of the E-NMOS in three modes.

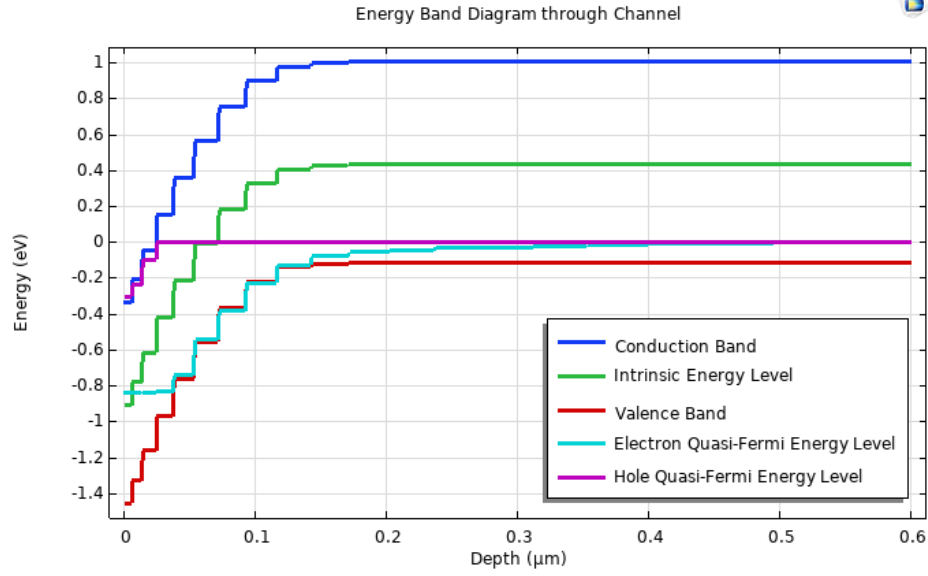


Figure 8: Band diagram through the gate and channel of the E-NMOS for $V_G = 4\text{ V}$, $V_D = 5\text{ V}$. Depth is distance beneath the gate.

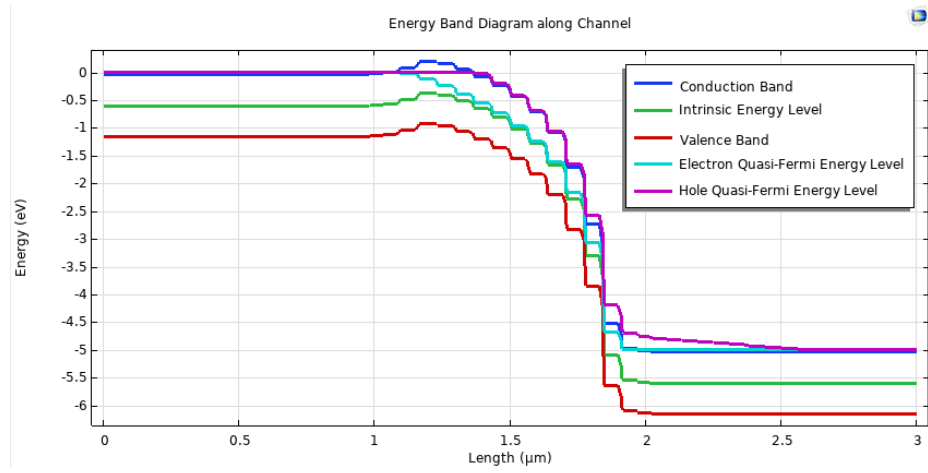


Figure 9: Band diagram along the channel of the E-NMOS for $V_G = 4\text{ V}$, $V_D = 5\text{ V}$. Length is distance from the boundary at the source and extends to the boundary at the drain.

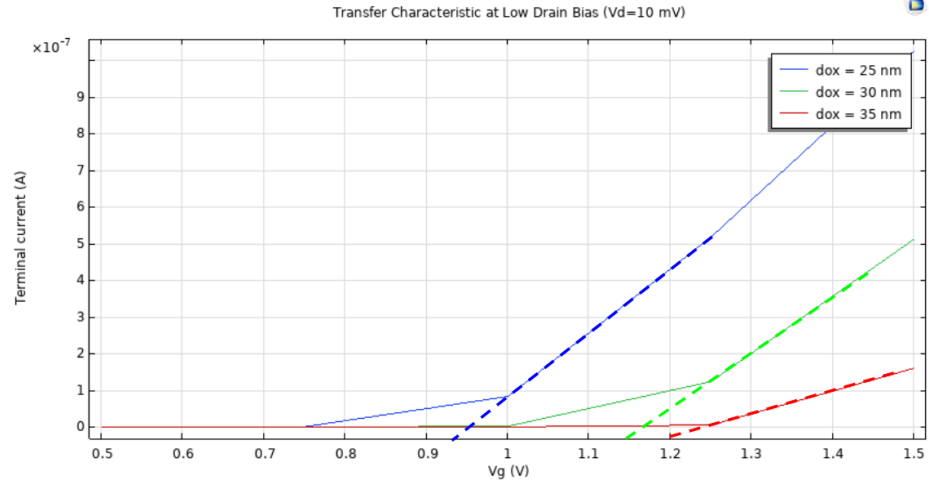


Figure 10: Plot of the IV transfer characteristics of the E-NMOS for three different oxide thicknesses d_{ox} . Dotted lines are extrapolated from the linear regime to estimate threshold voltage.

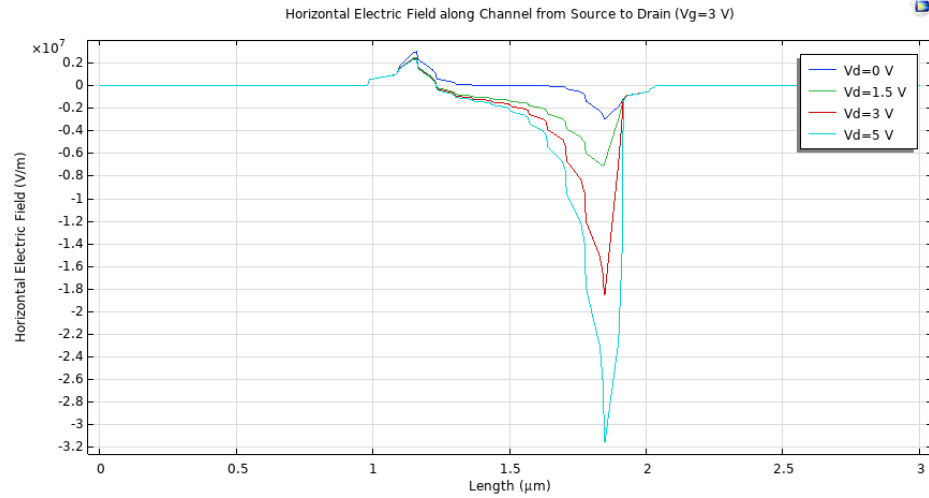


Figure 11: Plot of the horizontal electric field along the channel of the E-NMOS at $V_G = 3$ V for a range of V_D .

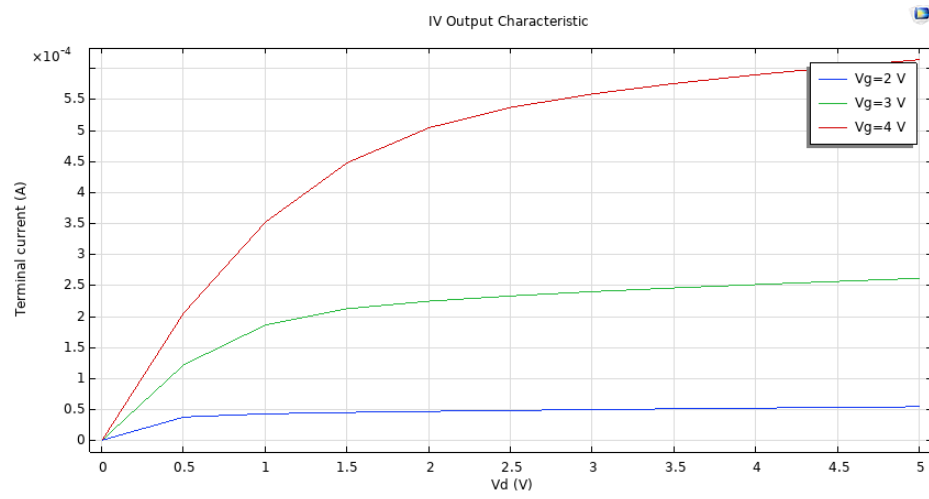


Figure 12: Plot of the IV output characteristics of the E-NMOS for three gate voltages V_G and $V_D = 10$ mV.

3.2 D-NMOS

At equilibrium, the D-NMOS is inverted due to n-type doping in the channel, and using a negative gate voltage $V_G < V_T < 0$ we may deplete the channel as per Figure 13. The band diagram along the channel (Figure 14) follows the same physics as in the E-NMOS, with the gate voltage translated downwards.

In Figure 15, one can see the effect of channel dopant concentration on the transfer characteristic and threshold voltage. It is intuitive that a stronger doping ($10^{17.75}$ to 10^{18} cm^{-3}) requires a stronger gate voltage to deplete (-0.5 to -1.8 V), and this is consistent with the simulation.

In the depletion regime, the IV output characteristic is the same as for the E-NMOS with a sub-threshold curve ($V_G = -2 \text{ V}$), and inversion for $V_G = -1 \text{ V}$ showing linear and saturation regions with output resistance $r_0 = 10^6 \Omega$.

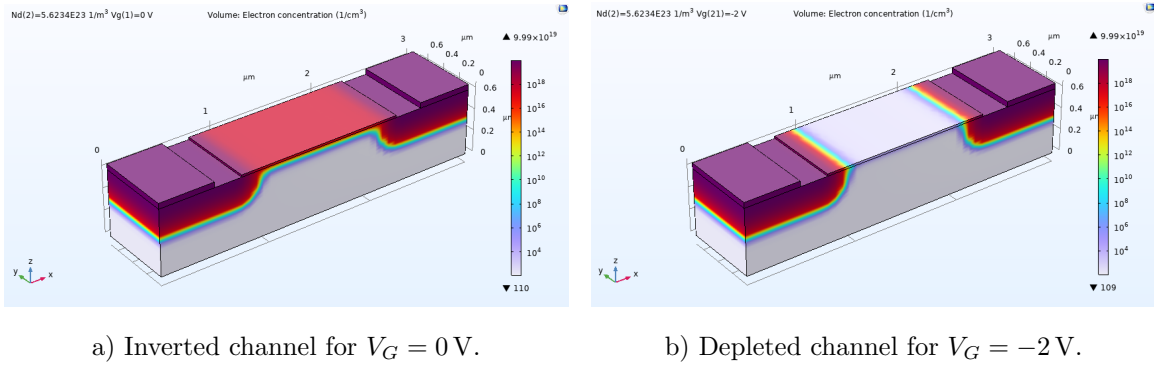


Figure 13: 3D plots of the electron concentration of the D-NMOS with drain voltage $V_D = 10 \text{ mV}$.

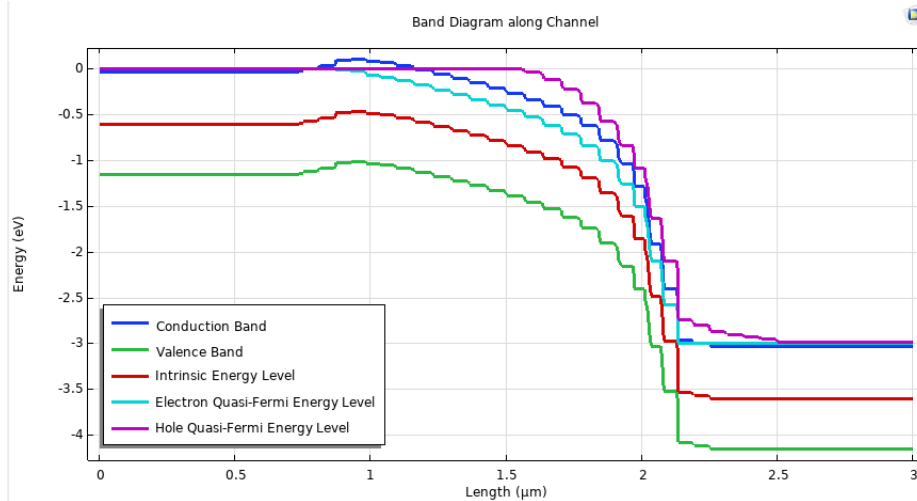


Figure 14: Band diagram along the channel of the D-NMOS for $V_G = 0 \text{ V}$, $V_D = 3 \text{ V}$. Length is distance from the boundary at the source and extends to the boundary at the drain.

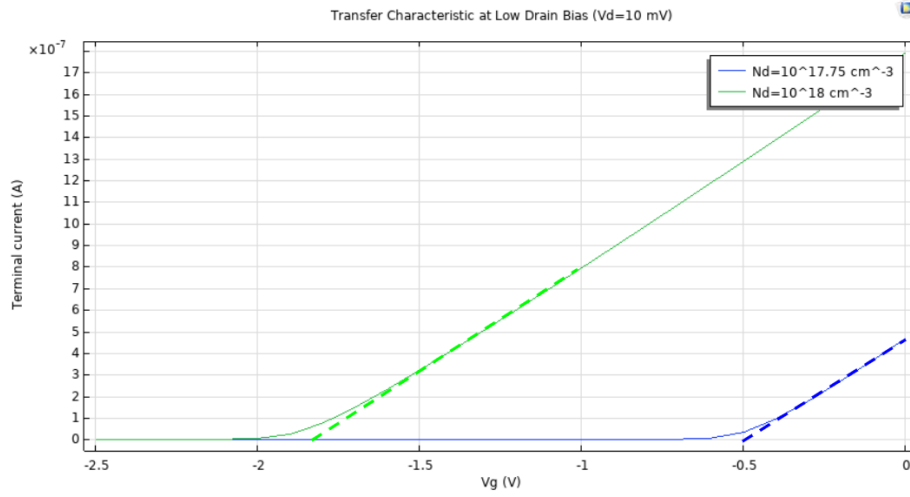


Figure 15: Plot of the IV transfer characteristics of the D-NMOS for two different n-type dopant concentrations N_d . Dotted lines are extrapolated from the linear regime to estimate cutoff voltage.

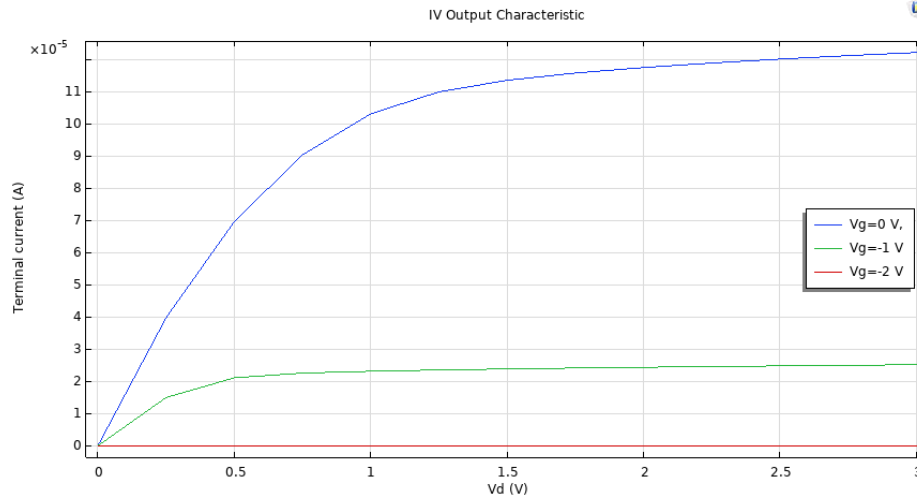


Figure 16: Plot of the IV output characteristics of the NMOS in depletion mode for three gate voltages V_G and $V_D = 10$ mV.

3.3 E-PMOS

The E-PMOS bares a strong resemblance to the E-NMOS, except that the holes are the majority current carrier, and that comes with some minus signs. We instead apply a negative gate voltage to invert the n-doped channel into a p-type channel in which holes can flow from source to drain (Figure 17). The vertical band diagram (Figure 18) bends upwards towards the gate, which can be explained by the negative electrostatic potential and opposite electric field of the E-NMOS, or instead by considering that the valence band is to the hole as the conduction is to the electron, and through this we can associate the energy of an unoccupied state to be $-E$, and appreciate the reflection of this band diagram through $E = 0$.

Through this heuristic we also understand Figure 19, where the hole quasi-Fermi energy increases $0 - 3\text{ eV}$ across the device due to a drain voltage $V_D = -3\text{ V}$ and holes move up the valence band with symmetric dynamics to the electrons. Since holes behave like positive charges, we therefore expect negative drain current I_D .

Figures 20 and 21 support this, with the same characteristic behaviour as the E-NMOS up to $I \rightarrow -I$, $V \rightarrow -V$. Our detailed analysis in that section transposes to here. We also verify that pinchoff occurs as expected in Figure 22 by the sharp peak in electric field.

When operating at -3 V , the on current is -1.5 A and the output resistance is $r_0 = 8 \times 10^5 \Omega$.

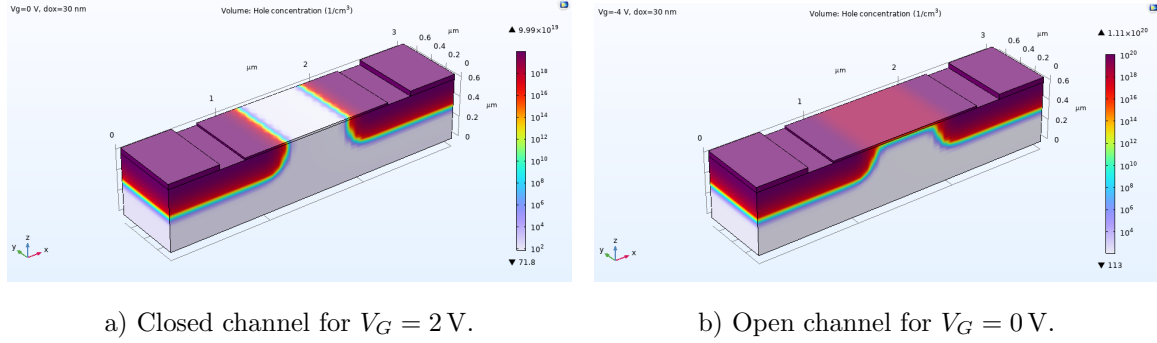


Figure 17: 3D plots of the hole concentration of the PMOS in enhancement mode with drain voltage $V_D = -10\text{ mV}$.

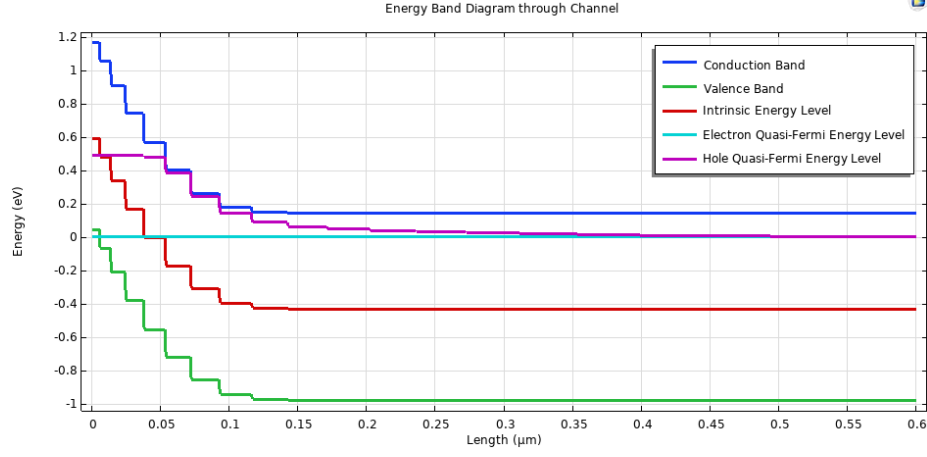


Figure 18: Band diagram through the gate and channel of the PMOS in enhancement mode for $V_G = -4\text{ V}$, $V_D = -3\text{ V}$. Depth is distance beneath the gate.

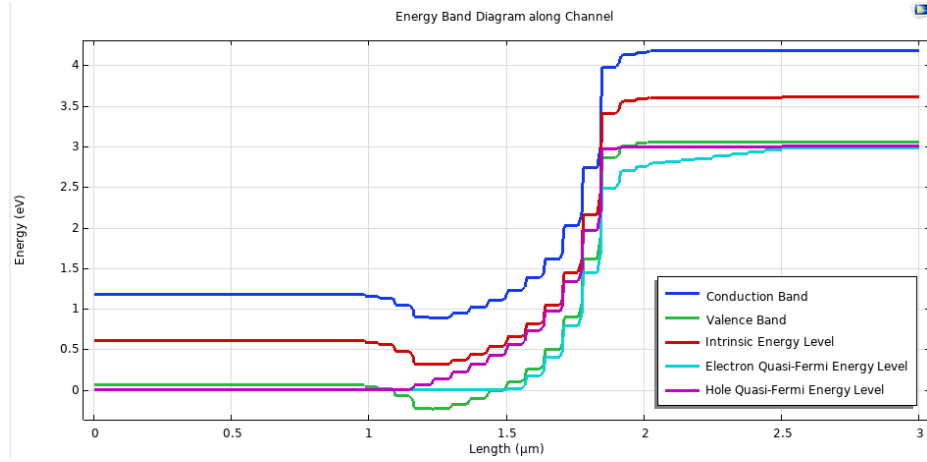


Figure 19: Band diagram along the channel of the PMOS in enhancement mode for $V_G = -4\text{ V}$, $V_D = -3\text{ V}$. Length is distance from the boundary at the source and extends to the boundary at the drain.

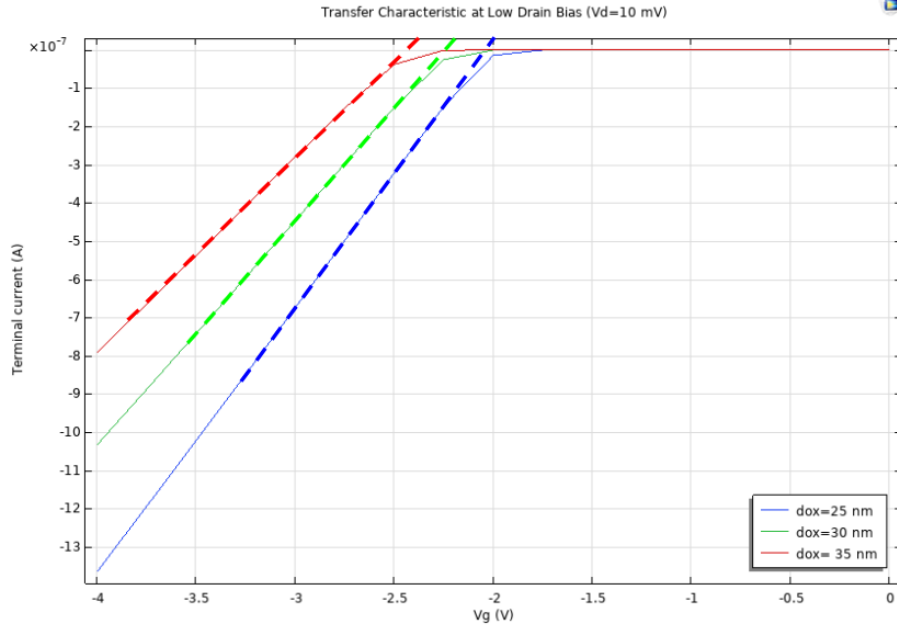


Figure 20: Plot of the IV transfer characteristics of the PMOS in enhancement mode for three different oxide thicknesses d_{ox} and $V_D = -10 \text{ mV}$. Dotted lines are extrapolated from the linear regime to estimate threshold voltage.

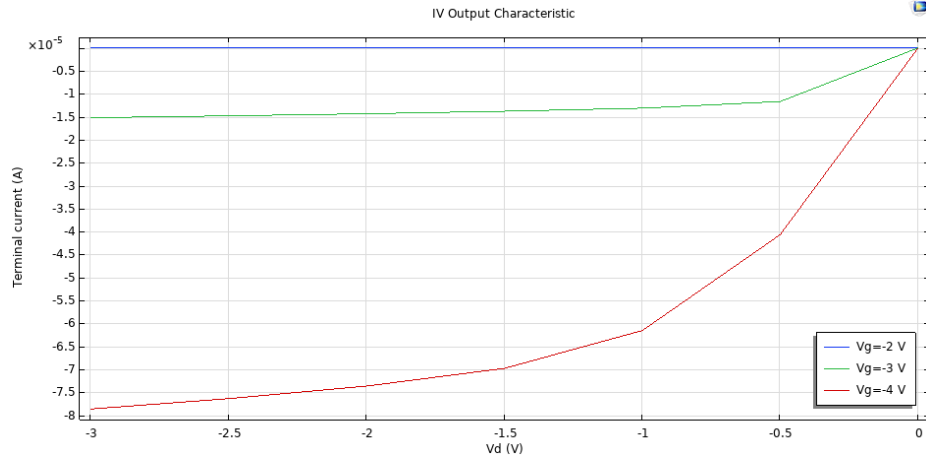


Figure 21: Plot of the IV output characteristics of the PMOS in enhancement mode for three gate voltages V_G .

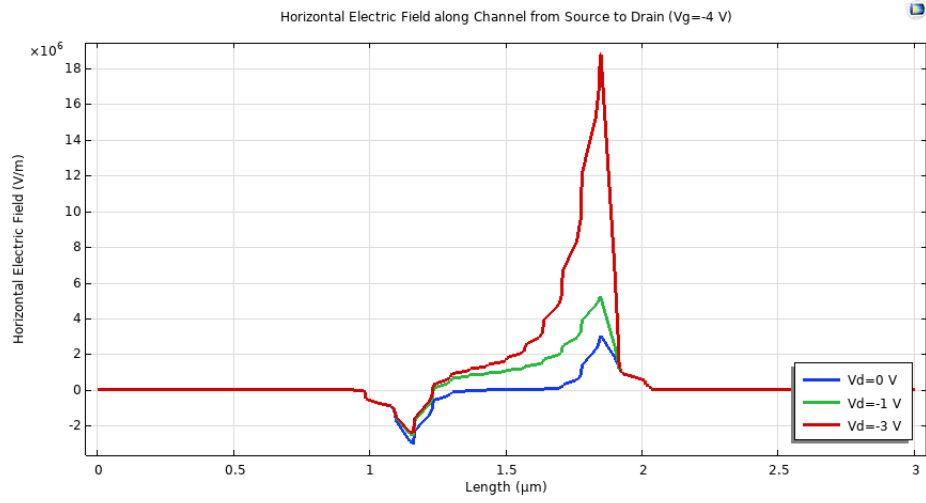


Figure 22: Plot of the horizontal electric field along the channel of the E-PMOS at $V_G = -4$ V for a range of V_D .

3.4 P-N Junction Diode

The analysis of the p-n junction diode is more straightforward than the MOSFET: it has one interface (the depletion region) and one anode voltage V_A to control (setting cathode voltage $V_C = 0$ V). In equilibrium (Figure 23), the bands are flat in the n and p-doped regions, with their respective raised and lowered bands. The drift and diffusion are in equilibrium:

$$\mathbf{J}_{\text{drift}} + \mathbf{J}_{\text{diffusion}} = 0. \quad (12)$$

Electrons diffuse from cathode to anode due to the concentration gradient [5], and drift from anode to cathode due to the Coulomb force discussed previously.

When we apply a forward bias $V_A > 0$ V, the energy bands in the p-type dopant are lowered such that the bands flatten (Figure 24), lowering the potential barrier and allowing minority charge carriers to drift across the depletion region, increasing the current (which reverses for high V_A). This allows a positive current to flow out the anode, given by the ideal p-n diode law.

$$I = I_S \left(\exp \left(\frac{qV_A}{nk_B T} \right) - 1 \right) \quad (13)$$

where I_S is the reverse saturation current and n is the ideality factor.

In our simulation, these are extracted from the IV characteristic and its semilog (Figures 25 and 26) as $I_S = 6 \times 10^{-19}$ A and $n = 1.01$. Note that $n = 1$ is an ideal diffusion-limited diode.

We now analyse the depletion region - donor electrons have recombined with acceptor atoms to form dopant ions, and these fixed charges have an internal electric field that opposes external change (Le Chatelier's Principle). As the voltage increases, the electrostatic potential barrier across the diode decreases (Figure 28), and the depletion width narrows in response to the reduced external electric field. This is demonstrated in Figure 29, where the width and intensity of the electric field fluctuation (i.e. the depletion region) decreases with voltage.

An extension of this is that a negative voltage (reverse bias) increases the energy barrier (Figure 30), electrostatic potential barrier (Figure 28), and depletion region width (Figure 29), thus drastically reducing output current. The small current that remains here is limited by minority carriers, and so current remains approximately constant (Figure 26).

Relevant performance metrics of the diode are the turn-on voltage $V_{ON} = V(I = 1 \text{ A}) = 0.9$ V, reverse leakage $I_{LEAK} = -4 \times 10^{-18}$ A and depletion width $W(V = 0) = 0.8 \mu\text{m}$.

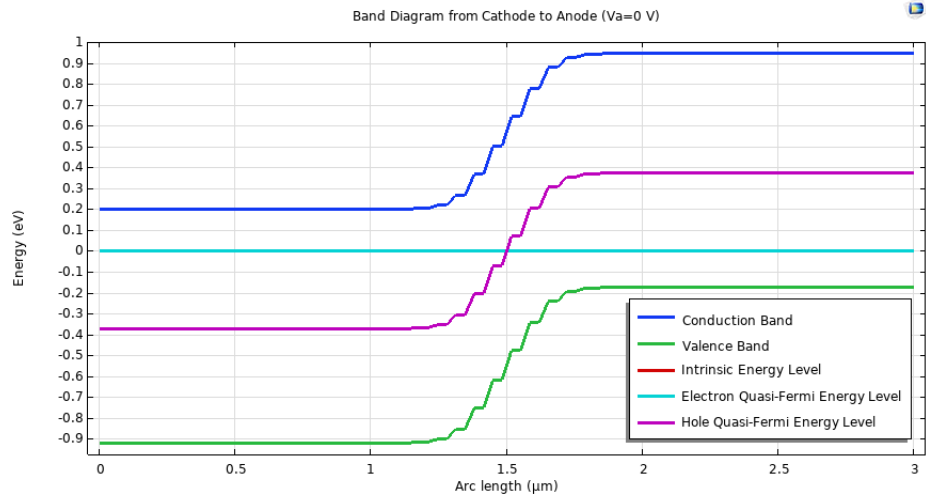


Figure 23: Band diagram of the p-n junction diode in equilibrium along the central axis from cathode to anode.

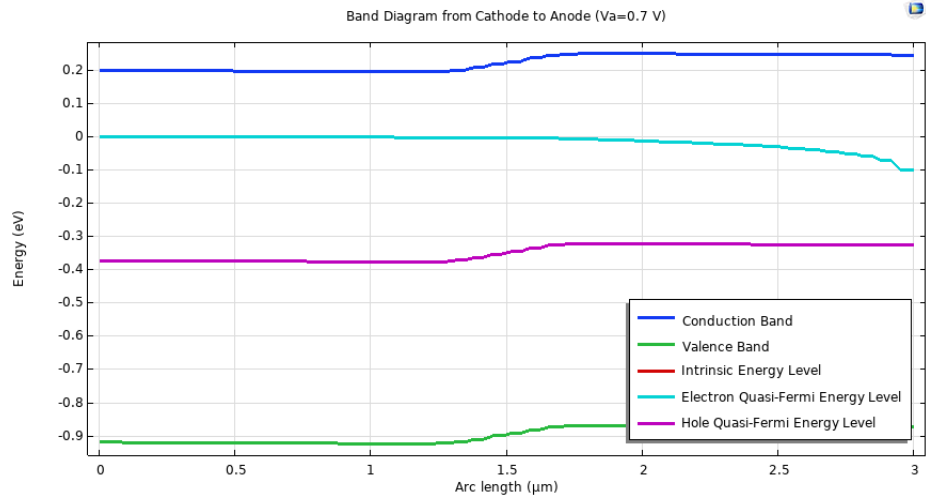


Figure 24: Band diagram of the p-n junction diode in forward bias along the central axis from cathode to anode.

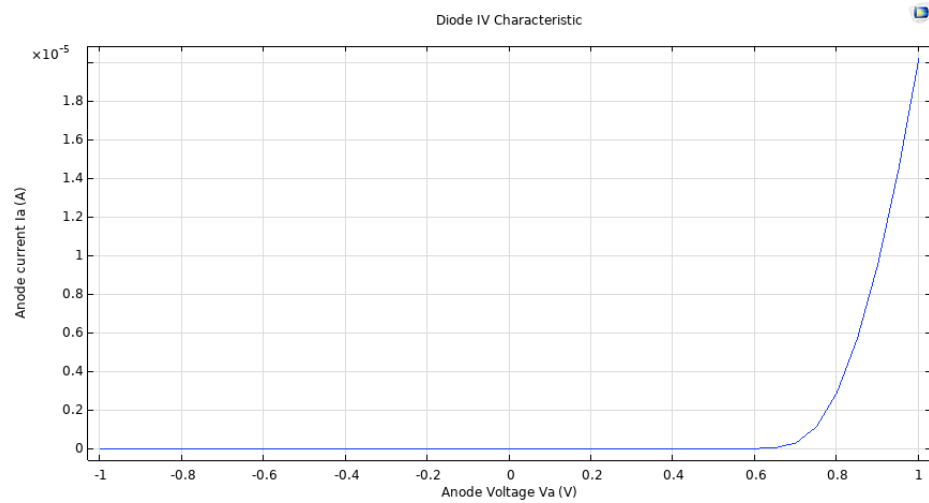


Figure 25: IV characteristic of the p-n junction diode along the central axis from cathode to anode.

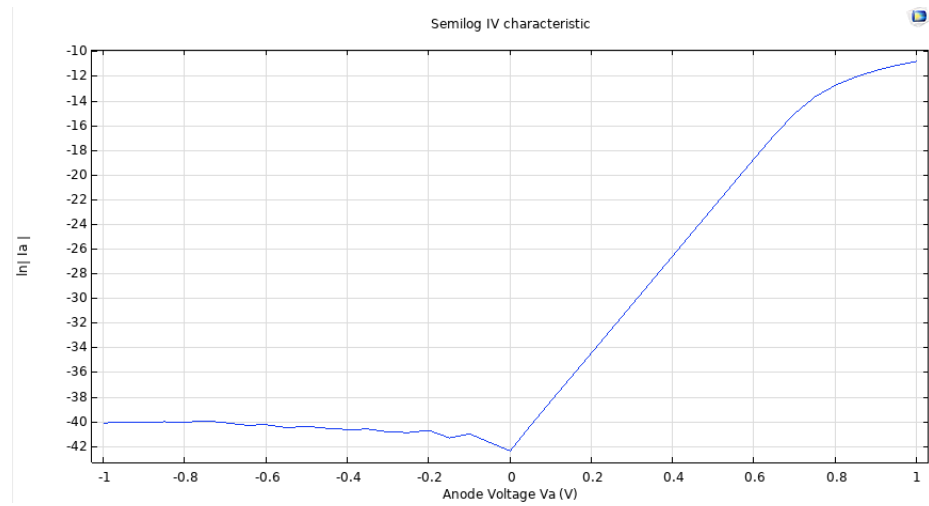


Figure 26: Semilog of IV characteristic of the p-n junction diode along the central axis from cathode to anode.

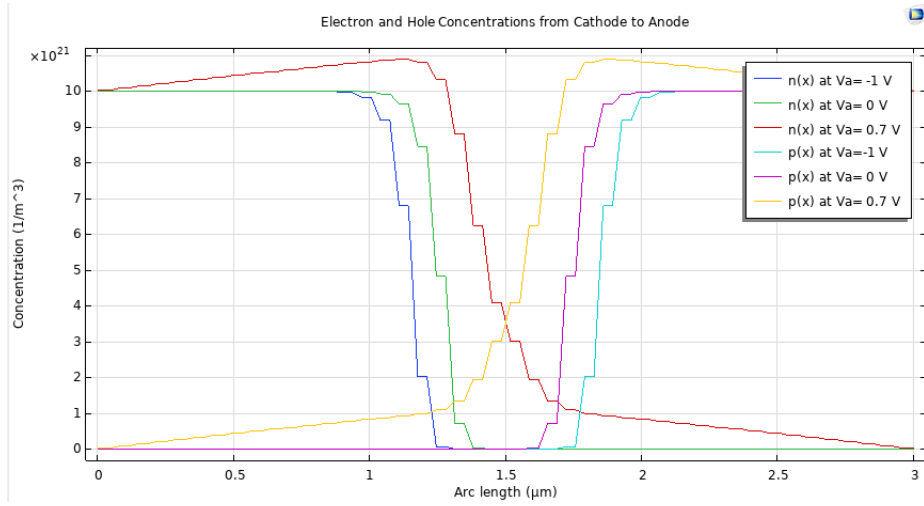


Figure 27: Charge carrier concentrations of the p-n junction diode along the central axis from cathode to anode.

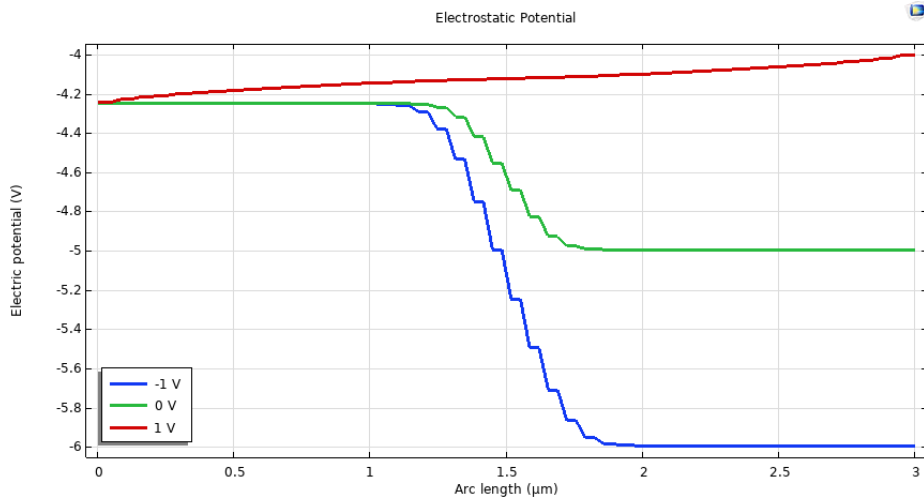


Figure 28: Electrostatic potential of the p-n junction diode for a range of anode voltages along the central axis from cathode to anode.

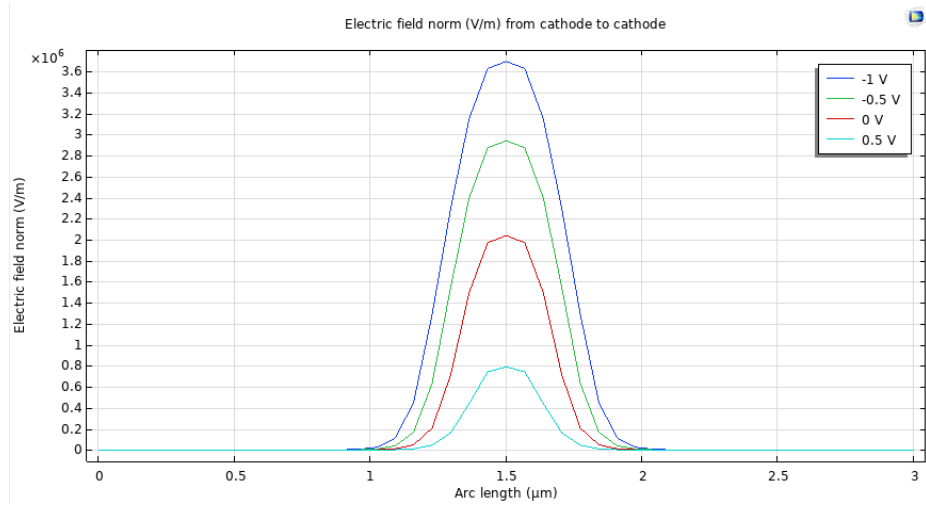


Figure 29: Electric field magnitude $|\mathbf{E}|$ for a range of voltages V_A from cathode to anode.

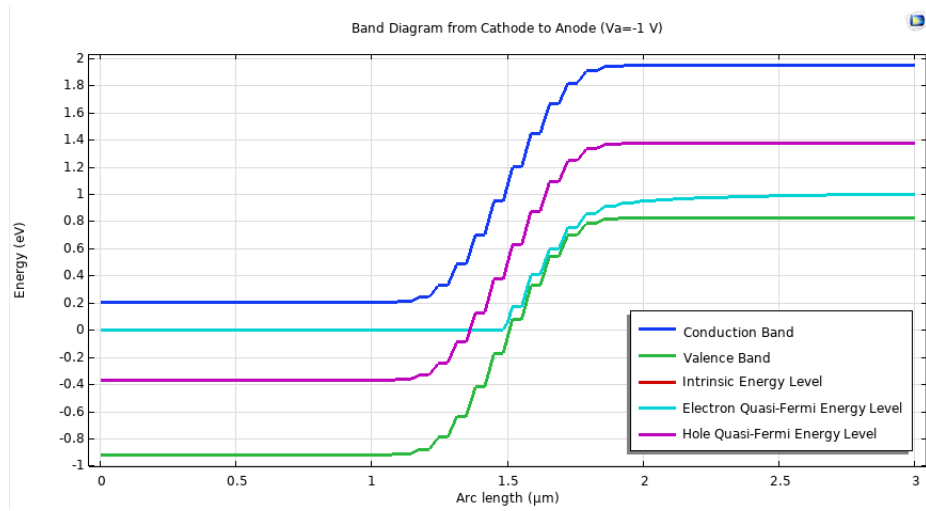


Figure 30: Band diagram of the p-n junction diode in reverse bias along the central axis from cathode to anode.

4 Discussion, Problem-Solving and Iteration

Following the tutorial, the simulations of the E-NMOS worked as intended. Swapping the dopants in the E-PMOS, I was surprised to see that COMSOL struggled to converge on solutions in the inversion mode. Despite tweaking a great number of parameters (eg dopant concentrations, work function and oxide thickness), I could not resolve the issue until I resolved to ‘walk’ solutions from the equilibrium $V_G = V_D = 0$ towards the desired voltages, reusing my previous solution at each step. I wasted some time parametrising a walk in V_G - V_D space, yet in the end a precarious auxiliary sweep from $V_D = 0$ converged for gate voltages above threshold. The parameters most explored in the D-NMOS were the channel dopant concentration and junction depth, with the goal of achieving depletion for suitable gate voltage, this involved parametric sweeps and inspecting 3D electron concentration plots. The p-n junction diode was more straightforward, the convergence was very efficient and the biggest challenge was choosing a dopant concentration such that the turn-on voltage and depletion region were appropriate.

5 Reflection

My undergraduate background covered quantum mechanics and band structure (tight binding, nearly free electron models and scattering on a Bravais lattice, Bloch electron dynamics and the semi-classical equation of motion). However, I had no experience with semiconductors, doping or electronic devices (and hence COMSOL). As such, this project has served as a fantastic opportunity to learn the theory underpinning MOSFETs, p-n diodes and the technical skills to navigate COMSOL. This project has also been a lesson in patience and perseverance - the iterative process of grappling with the software was frustrating at times.

Two significant limitations to the simulation are that the model neglects quantum tunnelling and can only solve for a discretised mesh. Future improvements to the MOSFET could explore performance under varying geometry, and future p-n diode work could investigate the Zener effect.

6 Conclusion and Recommendations

To conclude, all four COMSOL simulations succeeded in fully displaying the expected device behaviour, and demonstrated the implications of design on performance. In the E-NMOS and E-PMOS, the oxide thickness was varied to control the threshold voltage, and in the D-NMOS the dopant concentration was varied with similar effect. We also evaluated operating voltage for amplifier applications, and found that a lower gate voltage gave higher output resistance. However, at higher operating voltage the MOSFETs had stronger on current, so could be able to switch faster for digital applications. I would therefore recommend optimising the operating voltage for the chosen application.

References

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ChatGPT was used to recommend reading materials [2], [5], and M. Lundstrom’s lecture series: *Essentials of MOSFETs*.